Certified Kubernetes Administrator (CKA) Exam Guide

The missing book by a Kubestronaut

Puru Tuladhar

Certified Kubernetes Administrator Exam Guide

Puru Tuladhar

Version 1.35, 2026-08-25

Table of Contents

Preface

This isn't a typical Kubernetes book. It doesn't attempt to cover Kubernetes in depth. Instead, it is a practical guide focused on helping you prepare for and pass the Certified Kubernetes Administrator (CKA) exam.

I have been where you are: studying for the CKA and trying to determine what really matters. A strong foundation in Kubernetes is essential, but passing the exam requires something more—hands-on, exam-style practice focused on the skills you will actually need under pressure.

This book is strictly aligned with the CKA exam curriculum. Its chapters follow the exam domains and competencies one by one, explaining what you need to know and, more importantly, what you need to be able to do to earn a good score. The exam is not primarily a test of how deeply you understand Kubernetes theory. It tests whether you can apply fundamental Kubernetes concepts to real clusters.

The greatest challenge is time. You have only 120 minutes--about two hours—to complete 15 to 20 performance-based tasks. This book will show you how to avoid common mistakes, recover when you get stuck, and develop a faster problem-solving workflow, all with one goal in mind: passing the exam.

I am Puru Tuladhar, Nepal's first Kubestronaut, and I have completed all five Kubernetes certifications: CKA, CKS, CKAD, KCNA, and KCSA. Everything in this book comes from someone who has prepared for and passed these exams.

I cannot guarantee that reading this book alone will be enough to pass the CKA. What I can say is that if you complete the hands-on practice outlined here, you will give yourself a strong chance of reaching the required passing score, which is currently 66%.

This is an opinionated book by design. It covers what I believe is essential for the exam and leaves out material that will not help you on exam day. My aim is to give you compact, practical knowledge that is difficult to find gathered in one place.

Chapter 1. Domains and Competencies

The Certified Kubernetes Administrator (CKA) certification is designed to verify that credential holders can perform the responsibilities of a Kubernetes administrator. In practice, Kubernetes administration is usually one part of a broader DevOps or Site Reliability Engineering (SRE) role rather than a standalone job title.

Kubernetes does not operate in isolation. Before preparing for the CKA, you should be comfortable with foundational DevOps skills such as working with Linux, using the command line, editing files in a terminal, and understanding basic networking and storage concepts.

During the exam, you must demonstrate proficiency in installing, configuring, and managing Kubernetes clusters. These are the official exam domains and competencies covered which this book is based on.

1.1. Domains and Weighting

Domain	Weight
Cluster Architecture, Installation and Configuration	25%
Workloads and Scheduling	15%
Services and Networking	20%
Storage	10%
Troubleshooting	30%
Total	**100%**

Chapter 2. Cluster Architecture, Installation and Configuration

Exam Objectives **Exam Weight: 25%**

1. Prepare the underlying infrastructure for installing a Kubernetes cluster
2. Create and manage Kubernetes clusters using kubeadm
3. Manage the lifecycle of Kubernetes clusters
4. Implement and configure a highly available control plane
5. Use Helm and Kustomize to install cluster components
6. Manage role-based access control (RBAC)
7. Understand extension interfaces (CNI, CSI, CRI, and others)
8. Understand CRDs and install and configure operators

2.1. Prepare the underlying infrastructure for Installing a Kubernetes Cluster

A cluster is a group of nodes, or machines, that work together. This is true whether you are building a database cluster, a storage cluster, or any other type of cluster. A Kubernetes cluster is no different.

In the context of the exam, underlying infrastructure refers to the Linux nodes that form the Kubernetes cluster.

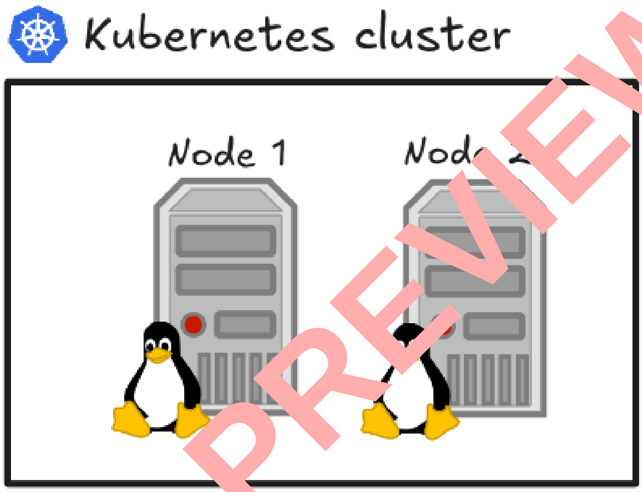

Figure 1. Kubernetes cluster composed of two Linux nodes

Kubernetes clusters can also use Windows, but Windows nodes are outside the scope of the exam.

The exam environment uses Ubuntu, a Debian-based Linux distribution, for Kubernetes cluster nodes. To keep our lab environment consistent with the exam, we will also use Ubuntu virtual machine for every Kubernetes node.

2.1.1. Virtual Machines

You can create the required virtual machines with VirtualBox, VMware, or any other virtualization platform available to you. This book uses Multipass because it is cross-platform and provides a straightforward way to create and manage Ubuntu virtual machines.

Allocate at least 2 GiB of RAM to each node and at least 2 vCPUs to control-plane node. These are the minimum hardware requirements documented for creating a cluster.

```
multipass launch -n control-plane -c 2 -m 2G 24.04 ①
multipass launch -n worker -c 1 -m 2G 24.04 ②
```

① Launch Ubuntu virtual machine and name it `control-plane`.